Integer points on a triple-induced curve

Problèmes ouverts — A. Dujella · 2026

Problem 4.8 (Dujella, 2026). Let $\{a, b, c\}$ be a Diophantine triple. Do all integer points on the elliptic curve $y^2 = (ax+1)(bx+1)(cx+1)$ (such that $x \neq -1$) satisfy the condition that $ax+1$, $bx+1$ and $cx+1$ are all perfect squares ?

Remark. Partial positive answers are known for the family of elliptic curves from Problem 4.9, and for families corresponding to triples of the form $\{1, 3, c\}$ (Dujella and Pethő, Publ. Math. Debrecen **56** (2000), 321–335) and $\{F_{2k}, F_{2k+2}, F_{2k+4}\}$ (Dujella, J. Théor. Nombres Bordeaux **13** (2001), 111–124).